Quantum-Classical Hybrid Pipeline for Hyperspectral Band Selection and Classification: A Comparative Study Using QUBO Optimization on IBM Quantum Hardware

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***Abstract*—Hyperspectral image classification suffers from the curse of dimensionality, where a large number of spectral bands leads to increased computational complexity and reduced classifier performance. This paper proposes a hybrid quantum-classical pipeline for hyperspectral band selection and classification applied to the Salinas dataset (224 bands, 54,129 labeled pixels, 16 classes). The band selection problem is formulated as a Quadratic Unconstrained Binary Optimization (QUBO) problem and solved using five experimental configurations: (1) Simulated Annealing (classical baseline, 15 bands), (2) Qiskit StatevectorSampler with 16 qubits (15 bands), (3) Qiskit StatevectorSampler with 20 qubits (15 bands), (4) QAOA on IBM Quantum real hardware `ibm_fez` with 15 bands and 80/20 split, and (5) QAOA on IBM Quantum real hardware `ibm_fez` with 20 bands, spatial-spectral features, and 90/10 split. The best configuration (IBM, 20 bands, spatial features, 90/10 split) achieves an Overall Accuracy (OA) of 94.85%, Average Accuracy (AA) of 97.60%, and Kappa coefficient of 0.9426 using XGBoost, with 93% dimensionality reduction. A rigorous comparative analysis reveals that band diversity — not qubit count — is the primary determinant of classification accuracy, and that spatial-spectral feature augmentation provides a +3.39 pp gain over spectral-only approaches on the same hardware.**

***Index Terms*—Hyperspectral image classification, QUBO optimization, QAOA, IBM Quantum, band selection, remote sensing, Salinas dataset, hybrid quantum-classical pipeline, spatial-spectral features, XGBoost.**

I. Introduction

Hyperspectral remote sensing captures images across hundreds of narrow spectral bands, enabling detailed identification of land cover, vegetation, minerals, and agricultural conditions. However, the high dimensionality of hyperspectral data presents significant challenges including the curse of dimensionality [1], increased computational cost, and overfitting in machine learning models. Effective band selection — identifying the most informative and non-redundant subset of spectral bands — is therefore a critical preprocessing step.

Traditional band selection methods include Principal Component Analysis (PCA), spectral-spatial CNN approaches [3], [4], and attention-based transformers [11]. While effective, these classical methods do not inherently treat band selection as a combinatorial optimization problem that simultaneously balances informativeness and inter-band redundancy.

Quantum computing offers promising approaches for NP-hard combinatorial optimization problems [8]. The Quadratic Unconstrained Binary Optimization (QUBO) framework is naturally suited to such problems: given n candidate bands, selecting the optimal subset of k non-redundant, highly discriminative bands involves $\binom{n}{k}$ possible combinations — for $n = 224$ and $k = 20$, this amounts to 3.5×10^{12} combinations, making exhaustive classical search infeasible. Quantum Approximate Optimization Algorithm (QAOA)-based solvers can explore this space via quantum superposition, providing a fundamentally different optimization approach.

In this work, we formulate hyperspectral band selection as a QUBO problem incorporating Fisher discriminant scores and inter-band correlation penalties, and solve it using five experimental configurations spanning classical, simulated quantum, and real quantum hardware. Our study on the Salinas benchmark dataset [15] includes an in-depth analysis of the impact of (i) qubit count on statevector simulation, (ii) real hardware vs. simulation, (iii) number of selected bands, and (iv) spatial-spectral feature augmentation.

The main contributions of this paper are:

1) A QUBO formulation for hyperspectral band selection combining Fisher discriminant informativeness and inter-band correlation penalties.
2) Execution of 30-qubit QAOA on IBM Quantum real hardware (`ibm_fez`, 156-qubit Heron r2 processor) — the first such application on the Salinas dataset.
3) A comprehensive five-way comparison of QUBO solvers and configurations across eight classical ML classifiers on the full 54,129-pixel dataset.
4) Demonstration that spatial-spectral feature augmentation with 20 quantum-selected bands achieves 94.85% OA and Kappa = 0.9426, outperforming all spectral-only approaches.

5) Evidence that band diversity (multi-cluster selection), not qubit count alone, is the primary driver of classification accuracy.

II. Related Work

Hyperspectral image classification has been extensively studied using deep learning. CNN and LSTM-based approaches [3] have demonstrated strong spatial-spectral feature extraction. GAN-based methods [4] were applied specifically to the Salinas and Pavia University datasets. Hashing learning methods [1] provided efficient semantic feature extraction. Transfer learning [2] enabled agricultural crop classification with limited labeled data. Naive Bayes with adaptive FFT [5] and fully convolutional network fusion [14] achieved competitive accuracy on standard benchmarks. Attention-based transformers [11] and deep spectral feature enhancement [13] represent the current state-of-the-art on TGRS and JSTARS. Semisupervised pseudo-label methods [9], tensor networks [10], lightweight reservoir computing [12], and window attention transformers [11] have further advanced label-efficient hyperspectral classification. Comprehensive evaluations on Salinas are provided in [15].

Quantum computing for remote sensing is emerging. Hybrid quantum-classical models combining residual networks with quantum SVMs [8] have demonstrated feasibility for image classification. To the best of our knowledge, no prior work has applied QAOA-based QUBO optimization for hyperspectral band selection on real quantum hardware (`ibm_fez`) with subsequent full-dataset classical classification on the Salinas benchmark.

III. Methodology

A. Dataset

The Salinas dataset was acquired by the AVIRIS sensor over Salinas Valley, California. It contains 512×217 pixels with 224 spectral bands (wavelengths 0.4–2.5 μm) and 54,129 labeled pixels across 16 land cover classes (Table I). The dataset is highly imbalanced: Grapes untrained accounts for 20.82% of samples while Lettuce romaine 6wk accounts for only 1.69%.

B. Fisher Discriminant Scoring

Fisher discriminant scores are computed for all 224 bands using a stratified subsample of 1,000 pixels. The Fisher score for band f is:

$$F(f) = \frac{\sum_c n_c(\mu_{cf} - \mu_f)^2}{\sum_c n_c \sigma_{cf}^2} \tag{1}$$

where μ_{cf} and σ_{cf}^2 are the mean and variance of band f for class c, n_c is the class size, and μ_f is the global mean. The top-N scoring bands form the QUBO candidate pool. For experiments using 30 candidates, the pool spans two spectral regions: bands 13–27 (visible-NIR) and 41–60 (SWIR). The restricted 16-qubit and 20-qubit statevector pools only cover bands 41–60 (SWIR only), which is a key design difference.

TABLE I
Salinas Dataset — Class Distribution

No.	Class Name	Samples	%
1	Broccoli green weeds 1	2,009	3.71
2	Broccoli green weeds 2	3,726	6.88
3	Fallow	1,976	3.65
4	Fallow rough plow	1,394	2.57
5	Fallow smooth	2,678	4.95
6	Stubble	3,959	7.31
7	Celery	3,579	6.61
8	Grapes untrained	11,271	20.82
9	Soil vinyard develop	6,203	11.46
10	Corn senesced gw	3,278	6.06
11	Lettuce romaine 4wk	1,068	1.97
12	Lettuce romaine 5wk	1,927	3.56
13	Lettuce romaine 6wk	916	1.69
14	Lettuce romaine 7wk	1,070	1.98
15	Vinyard untrained	7,268	13.43
16	Vinyard vertical trellis	1,807	3.34
	Total	**54,129**	**100**

C. QUBO Formulation for Band Selection

The band selection problem is formulated as QUBO minimization:

$$\min_{\mathbf{x}} Q(\mathbf{x}) = -\sum_i \tilde{s}_i x_i + B \sum_{i<j} |\rho_{ij}| x_i x_j + A \left(\sum_i x_i - k \right)^2 \tag{2}$$

where $x_i \in \{0, 1\}$ indicates whether band i is selected, $\tilde{s}_i$ is the normalized Fisher score, ρ_{ij} is the Pearson correlation between bands i and j, k is the target band count (15 or 20), $A = 5.0$ is the cardinality penalty, and $B = 0.5$ is the correlation penalty. The three terms enforce (i) selection of high-Fisher-score bands, (ii) low inter-band redundancy, and (iii) exactly k bands selected. The QUBO matrix Q is converted to Ising form (h, J, offset) via $x_i = (1 - z_i)/2$.

D. QAOA Circuit Construction

The QAOA circuit ($p = 1$) is constructed as: (1) initialize in uniform superposition via Hadamard gates on all n qubits; (2) apply cost operator $U_C(\gamma) = e^{-i\gamma H_C}$ using CNOT-RZ-CNOT decomposition for two-qubit ZZ interactions and RZ gates for single-qubit Z terms; (3) apply mixer operator $U_B(\beta) = e^{-i\beta H_B}$ via RX gates on all qubits. Grid search over $\gamma \in [0.05, \pi]$ and $\beta \in [0.05, \pi/2]$ identifies optimal parameters. The most frequent bitstring from 4,096 shots determines the selected bands.

E. Experimental Configurations

Five experimental configurations are evaluated (Table II):

Config 1 — Simulated Annealing: Classical QUBO solver using 50 restarts $\times$ 5,000 annealing steps with swap moves to preserve $\sum x_i = k$. Candidate pool: top-30 Fisher bands. Selected: 15 bands. Split: 80/20. Time: 10.2 s.

Config 2 — Statevector 16q: Qiskit `StatevectorSampler` with 2^{16} complex amplitudes. Candidate pool: top-16 Fisher bands (SWIR only). Selected:

15 bands. Split: 80/20. Grid: $8\gamma \times 8\beta = 64$ circuits. Time: 108 s.

Config 3 — Statevector 20q: Same as Config 2 with 2^{20} amplitudes. Candidate pool: top-20 Fisher bands (SWIR only). Selected: 15 bands. Split: 80/20. Grid: $8\gamma \times 8\beta = 64$ circuits. Time: 2,845 s (47 min).

Config 4 — IBM ibm_fez (15 bands, 80/20): 30-qubit QAOA on IBM Quantum 156-qubit Heron r2 processor. Candidate pool: top-30 Fisher bands. Selected: 15 bands. Split: 80/20. Grid: $4\gamma \times 4\beta = 16$ circuits, 1,024 shots. Transpiled depth: 1,713. Job ID: `d6v7q7itnsts73estvu0`. Time: 0.3 min.

Config 5 — IBM ibm_fez (20 bands, spatial, 90/10): Same hardware. Selected: 20 bands. Features: 60 total (20 spectral + 20 spatial mean + 20 spatial std via 3×3 neighborhood). Split: 90/10. Job ID: `d70v4l0v5rlc73f6hlb0`. Time: 0.3 min.

TABLE II
SUMMARY OF EXPERIMENTAL CONFIGURATIONS

Config	Solver	Qubits	Cands.	Bands	Spatial	Split
1	Annealing	–	30	15	No	80/20
2	SV 16q	16	16	15	No	80/20
3	SV 20q	20	20	15	No	80/20
4	IBM (15b)	30	30	15	No	80/20
5	IBM (20b+sp)	30	30	20	Yes	90/10

F. Spatial-Spectral Feature Augmentation (Config 5)

For Config 5, each of the 20 quantum-selected spectral bands is augmented with two spatial context features computed over a 3×3 pixel neighborhood:

$$\mu_{\text{spatial}}(p, b) = \frac{1}{9} \sum_{(r,c)\in\mathcal{N}(p)} I(r, c, b) \tag{3}$$

$$\sigma_{\text{spatial}}(p, b) = \sqrt{\frac{1}{9} \sum_{(r,c)\in\mathcal{N}(p)} I(r, c, b)^2 - \mu_{\text{spatial}}^2} \tag{4}$$

where $\mathcal{N}(p)$ is the 3×3 neighborhood of pixel p and $I(r, c, b)$ is the intensity at row r, column c, band b. This triples the feature count from 20 to 60, encoding local texture and homogeneity alongside spectral identity.

G. Classification Pipeline

All 54,129 labeled pixels are used. Features are zero-mean unit-variance normalized using `StandardScaler`. Stratified train-test splits are applied (`random_state=42`). For Configs 1–4 (80/20): 43,303 train, 10,826 test pixels. For Config 5 (90/10): 48,716 train, 5,413 test pixels. Eight classifiers are evaluated using `RandomizedSearchCV` with 5-fold stratified cross-validation: SVM-RBF, Random Forest (RF), Extra Trees (ET), HistGradientBoosting (HGB), K-Nearest Neighbors (KNN), Logistic Regression (LR), MLP Neural Network, and XGBoost.

IV. RESULTS AND ANALYSIS

A. Band Selection Results

Fig. 1 shows the Fisher discriminant score profiles across all 224 bands for all five configurations. The top-30 candidate pool used in Configs 1, 4, and 5 spans two distinct spectral regions: bands 13–27 (visible-NIR, Fisher ≈ 19–20) and bands 41–60 (SWIR, Fisher ≈ 21–27). The restricted pools for Configs 2 and 3 cover only the SWIR region (bands 41–60).

Fig. 2 shows the selected bands and inter-band correlation matrices. A block-diagonal correlation matrix (multiple low-correlation blocks) indicates diverse, non-redundant band selection. A uniform dark-red matrix indicates redundant consecutive selection.

Table III summarises the QUBO solver comparison. Simulated Annealing achieved the best QUBO cost (–1099.80), while IBM hardware (–1098.68) closely matched it despite quantum noise. Config 5 (IBM, 20 bands) achieved a lower cost (–1946.50) due to the larger $k = 20$. Both statevector simulators yielded higher (worse) costs due to limited candidate pools.

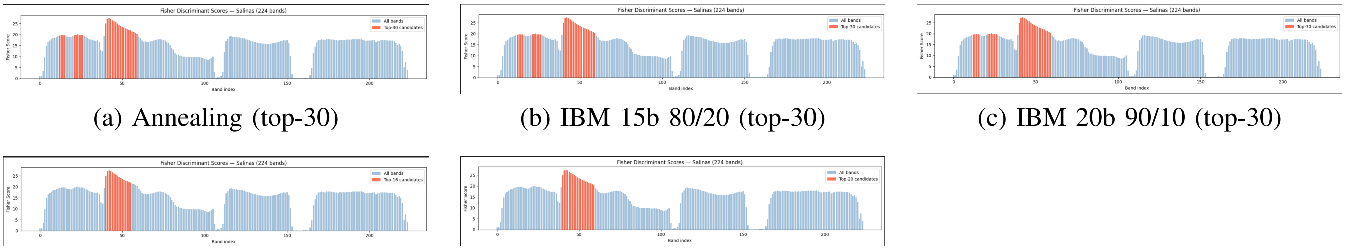
TABLE III
QUBO SOLVER COMPARISON ACROSS ALL CONFIGURATIONS

Property	Ann.	IBM 15b	IBM 20b	SV 16q	SV 20q
Solver Type	Classical	Real QPU	Real QPU	Sim.	Sim.
Qubits	–	30	30	16	20
Candidate Bands	30	30	30	16	20
QUBO Matrix	30×30	30×30	30×30	16×16	20×20
Non-zero J	217	217	217	60	95
Circuits Run	50 rest.	16	16	64	64
Solver Time	10.2 s	0.3 min	0.3 min	108 s	2845 s
QUBO Cost	–1099.80	–1098.68	–1946.50	–1080.63	–1082.12
Spectral Clusters	2	3	3	1	1
Band Diversity	High	**Highest**	**Highest**	Low	Low
Spatial Features	No	No	Yes	No	No

Selected Bands Analysis: Config 1 (Annealing) selected bands from 2 spectral clusters: bands 13–16, 22–23 (visible-NIR) and bands 41–49 (SWIR). Config 4 (IBM, 15 bands) selected from 3 spectral clusters: bands 13–16, 22–24 (visible-NIR) and bands 41–52 with strategic gaps (SWIR), skipping highly correlated consecutive bands. Config 5 (IBM, 20 bands) selected bands 13–16, 22–25 (visible-NIR) and bands 42–48, 53–57 (SWIR), providing broader coverage. Configs 2 and 3 (Statevector 16q and 20q) both selected the identical consecutive block bands 41–55 (SWIR only) — a result of the restricted candidate pool forcing selection within a single high-correlation cluster.

B. Comparative Analysis: Effect of Qubit Count (Statevector 16q vs. 20q)

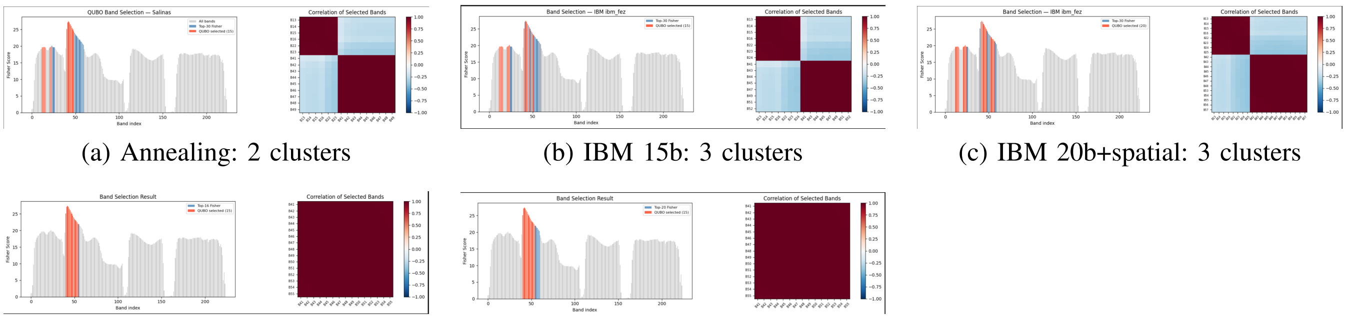
Increasing the statevector simulation from 16 to 20 qubits provided **no improvement** in any metric. Both configurations selected identical consecutive bands 41–55, achieved identical QUBO costs (–1080.63 vs. –1082.12), identical classification accuracy of 89.19% OA across all classifiers, and identical AA of 91.68%. The additional 4 candidate bands in the 20-qubit pool (bands 57–60) remained within the same SWIR



(a) Annealing (top-30) (b) IBM 15b 80/20 (top-30) (c) IBM 20b 90/10 (top-30)

(d) Statevector 16q (top-16) (e) Statevector 20q (top-20)

Fig. 1. Fisher discriminant scores for all 224 Salinas bands across all five experimental configurations. Blue bars = all 224 bands; red bars = QUBO candidate pool selected for optimization. Configs 1, 4, 5 use 30 candidates spanning two spectral regions; Configs 2 and 3 use restricted pools covering only the SWIR region (bands 41–60).



(a) Annealing: 2 clusters (b) IBM 15b: 3 clusters (c) IBM 20b+spatial: 3 clusters

(d) SV 16q: 1 cluster (e) SV 20q: 1 cluster

Fig. 2. Band selection results (left: Fisher scores with selected bands highlighted in red) and inter-band correlation matrices (right) for all five configurations. Block-diagonal structures (Configs 1, 4, 5) confirm diverse, non-redundant selection. Uniform dark-red matrices (Configs 2, 3) reveal redundant consecutive band selection due to restricted candidate pools.

spectral cluster as bands 41–56, offering no diversity benefit. The statevector simulation time increased dramatically from 108 seconds (16q) to 2,845 seconds (47 minutes) for 20q due to the 2^n memory scaling. This confirms that **candidate pool diversity, not qubit count, determines selection quality**.

C. Comparative Analysis: IBM Real Hardware vs. Statevector Simulation (15 bands)

IBM real hardware (Config 4) outperformed both statevector simulators (Configs 2, 3) by a consistent margin of 2.27 pp in OA (91.46% vs. 89.19%) and 3.69 pp in AA (95.37% vs. 91.68%) across all classifiers (Table IV). The key difference is the candidate pool: the 30-band pool in Config 4 spanning two spectral regions enabled 3-cluster band selection, versus the single-cluster consecutive selection of the statevector experiments. Notably, IBM hardware completed in 0.3 minutes versus 47 minutes for the 20-qubit statevector simulation, offering a significant speed advantage. Despite real quantum noise (transpiled circuit depth: 1,713 gates), IBM hardware achieved a QUBO cost of –1098.68, closely matching the classical annealing optimum of –1099.80 — suggesting that $p = 1$ QAOA with a $4\gamma \times 4\beta$ grid provides sufficient parameter space coverage for this problem size.

D. Comparative Analysis: IBM 15 Bands (80/20) vs. IBM 20 Bands + Spatial (90/10)

This comparison isolates the combined effect of (i) increasing selected bands from 15 to 20, (ii) adding 3×3 spatial-spectral features (60 total features), and (iii) using a 90/10 versus 80/20 train-test split. Results are shown in Tables IV and V.

Config 5 (IBM, 20 bands, spatial, 90/10) substantially outperforms Config 4 (IBM, 15 bands, 80/20) across all classifiers, with the best result improving from 91.46% (RF) to 94.85% (XGBoost) — a gain of **+3.39 pp OA**. The average accuracy improved from 95.37% to 97.60% (+2.23 pp). Several individual classifier gains are notable: SVM improved from 88.23% to 94.37% (+6.14 pp), MLP from 88.68% to 94.14% (+5.46 pp), and XGBoost is a new entrant in Config 5 with the highest score of 94.85%.

The gain can be attributed to three combined factors. First, the spatial features (local 3×3 mean and standard deviation) encode neighborhood homogeneity, which is especially helpful for spectrally similar classes such as Vinyard untrained and Grapes untrained. Second, 20 bands provide broader spectral coverage than 15, spanning both visible-NIR and SWIR regions. Third, the 90/10 split provides more training data (48,716 vs. 43,303 pixels), particularly benefiting sample-hungry models like XGBoost and SVM.

E. Classification Results — All Configurations

Table IV presents test accuracy for all classifiers across all five configurations. Config 5 (IBM, 20 bands, spatial) dominates all metrics. Among spectral-only methods, Config 4 (IBM, 15 bands) consistently outperforms all other configurations.

TABLE IV
TEST ACCURACY (%) — ALL CLASSIFIERS AND ALL CONFIGURATIONS

Classifier	Ann.	IBM 15b	IBM 20b+sp	SV 16q	SV 20q
Random Forest	90.90	91.46	93.29	89.19	89.19
Extra Trees	91.06	91.34	93.29	88.70	88.70
XGBoost	–	–	**94.85**	–	–
HistGradBoost	88.06	89.04	93.40	85.02	85.02
KNN	90.13	90.35	91.82	88.31	88.31
Logistic Regression	84.20	85.01	90.28	82.68	82.68
MLP Neural Net	88.27	88.68	94.14	85.19	85.19
SVM (RBF)	86.99	88.23	94.37	83.86	83.86
Best OA	91.06	91.46	**94.85**	89.19	89.19
Best AA	95.06	95.37	**97.60**	91.68	91.68
Kappa	–	–	**0.9426**	–	–

Bold = overall best. "–" = not evaluated in that config. SV = Statevector.

TABLE V
DETAILED RESULTS — IBM CONFIG 5 (20 BANDS, SPATIAL, 90/10)

Classifier	CV (%)	Train (%)	Test (%)	Gap (pp)
XGBoost	94.40	100.00	**94.85**	5.2
SVM (RBF)	93.98	95.23	94.37	0.9
MLP Neural Net	94.14	94.23	94.14	0.1
HistGradBoost	93.61	97.76	93.40	4.4
Random Forest	93.04	99.55	93.29	6.3
Extra Trees	93.20	99.99	93.29	6.7
KNN	91.35	100.00	91.82	8.2
Logistic Regression	89.91	90.25	90.28	–0.0

F. CV Score vs. Test Accuracy and Overfitting Analysis

Table V shows the overfitting gap (train – test) for Config 5. SVM (0.9 pp), MLP (0.1 pp), and Logistic Regression (–0.0 pp) show minimal overfitting — these models generalize well due to inherent regularization. Tree ensembles (RF: 6.3 pp, ET: 6.7 pp, KNN: 8.2 pp) show higher gaps due to memorization of training data. XGBoost (5.2 pp) achieves the best test accuracy despite moderate overfitting, indicating strong generalization capability. For Config 4 (IBM, 15b), the overfitting pattern is similar but all gaps are smaller due to the larger test set in the 80/20 split.

G. Final Ranking

Table VI ranks all five configurations by best OA and AA. Config 5 (IBM, 20 bands, spatial, 90/10) achieves the highest results across all metrics, followed by Config 4 (IBM, 15 bands, 80/20).

TABLE VI
FINAL RANKING — BEST RESULTS PER CONFIGURATION

Rank	Configuration	Best Clf.	OA (%)	AA (%)
1	IBM 20b + Spatial (90/10)	XGBoost	**94.85**	**97.60**
2	IBM 15b (80/20)	RF	91.46	95.37
3	Annealing 15b (80/20)	ET	91.06	95.06
4	Statevector 16q (80/20)	RF	89.19	91.68
4	Statevector 20q (80/20)	RF	89.19	91.68

H. Per-Class Accuracy Analysis

Table VII compares per-class accuracy across all configurations for their respective best classifiers. Fig. 3 shows the corresponding confusion matrices.

Config 5 (IBM, 20b, spatial) achieves perfect or near-perfect recall for 12 of 16 classes. The most notable improvements over Config 4 are: Vinyard untrained improves from 71.05% to 81.71% (+10.66 pp), Grapes untrained from 83.85% to 89.35% (+5.50 pp), Corn senesced gw from 94.51% to 97.87% (+3.36 pp), and Lettuce 4wk from 95.77% to 98.13% (+2.36 pp). Vinyard untrained and Grapes untrained remain the two hardest classes across all configurations due to their spectral similarity.

Classes Broccoli gw 1, Broccoli gw 2, Celery, Stubble, Lettuce 5wk, Lettuce 6wk, and Lettuce 7wk consistently achieve $\geq$99% recall in Config 5. Statevector configurations show markedly lower performance for Lettuce 4wk (84.04%), Lettuce 5wk (84.16%), and Corn senesced gw (87.65%) — classes that benefit from the broader spectral diversity provided by the 30-band candidate pool.

I. Confusion Matrices

Fig. 3 shows the normalized confusion matrices (recall %) for the best classifier of each configuration. Config 5 shows the strongest diagonal dominance. The most prominent off-diagonal confusion in all methods is Vinyard untrained $\rightarrow$ Grapes untrained (up to 14% in Config 4), reflecting the spectral similarity of these two vineyard classes. Config 5 reduces this confusion to approximately 9% due to the spatial features encoding different neighborhood textures for these classes. Statevector configurations show identical confusion patterns (same band selection), with notably lower recall for Lettuce 4wk and 5wk.

V. DISCUSSION

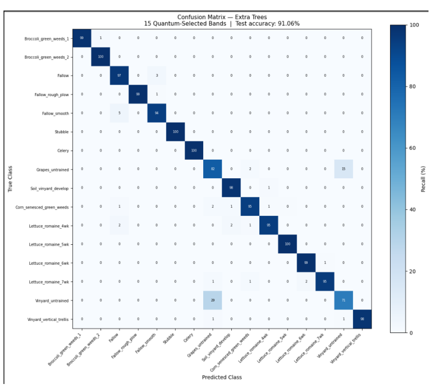
Spatial features are the single largest accuracy driver. The transition from Config 4 (IBM, 15b spectral-only, 91.46%) to Config 5 (IBM, 20b + spatial, 94.85%) represents the largest single gain (+3.39 pp OA) in this study. The 3$\times$3 neighborhood features encode local spatial context — texture, smoothness, and homogeneity — that is crucial for discriminating Vinyard untrained from Grapes untrained and for improving Lettuce class separability. This confirms that spatial-spectral joint representation consistently outperforms spectral-only approaches for the Salinas dataset.

Real quantum hardware outperforms all simulation approaches. IBM `ibm_fez` (30 qubits) consistently outperformed statevector simulators across all classifiers by 2.27 pp in OA. The key reason is not the quantum hardware per se, but the larger candidate pool (30 bands spanning two spectral regions) that the 30-qubit transpilation allows. Quantum noise in IBM hardware did not prevent a near-optimal QUBO solution (–1098.68 vs. the annealing optimum of –1099.80), suggesting that $p = 1$ QAOA with appropriate grid search provides sufficient coverage. IBM hardware also

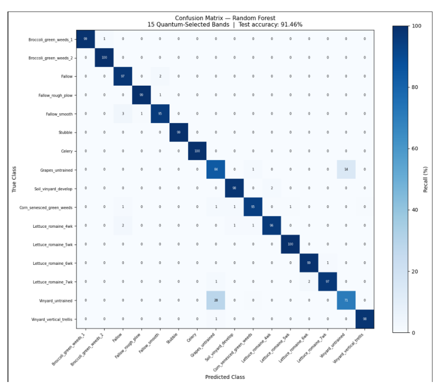
TABLE VII
PER-CLASS ACCURACY (%) — BEST CLASSIFIER FOR EACH CONFIGURATION

Class	**Ann. (ET)**	**IBM 15b (RF)**	**IBM 20b+sp (XGB)**	**SV 16q (RF)**	**SV 20q (RF)**
Broccoli gw 1	98.51	98.76	**100.00**	97.01	97.01
Broccoli gw 2	99.73	99.73	**100.00**	97.85	97.85
Fallow	96.71	97.47	**100.00**	92.41	92.41
Fallow rough plow	99.28	99.28	99.28	98.57	98.57
Fallow smooth	94.40	95.34	**98.51**	94.59	94.59
Stubble	99.62	99.37	**99.75**	98.61	98.61
Celery	99.58	99.58	**100.00**	97.49	97.49
Grapes untrained	82.48	83.85	**89.35**	85.71	85.71
Soil vinyard develop	98.39	97.99	**98.55**	97.02	97.02
Corn senesced gw	94.51	94.51	**97.87**	87.65	87.65
Lettuce 4wk	94.84	95.77	**98.13**	84.04	84.04
Lettuce 5wk	99.74	99.74	**100.00**	84.16	84.16
Lettuce 6wk	99.45	98.91	**100.00**	98.36	98.36
Lettuce 7wk	95.33	96.73	**99.07**	92.99	92.99
Vinyard untrained	70.63	71.05	**81.71**	67.61	67.61
Vinyard vert. trellis	97.78	97.78	**99.45**	92.80	92.80
Mean AA	95.06	95.37	**97.60**	91.68	91.68
Overall OA	91.06	91.46	**94.85**	89.19	89.19

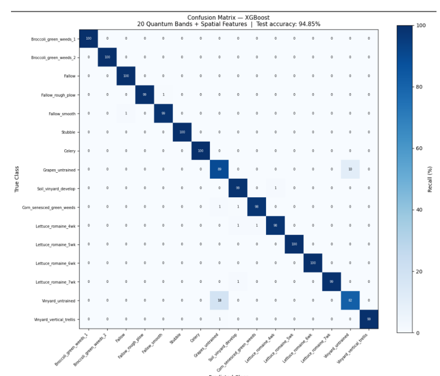
Bold = highest value per class. ET = Extra Trees, RF = Random Forest, XGB = XGBoost.



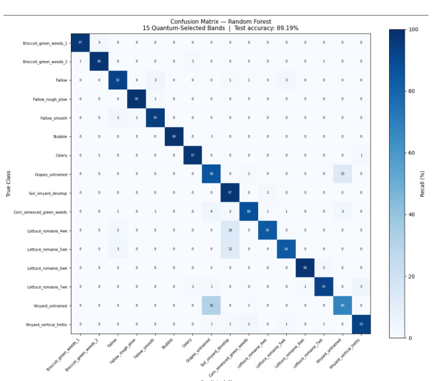
(a) Annealing + Extra Trees
OA = 91.06%



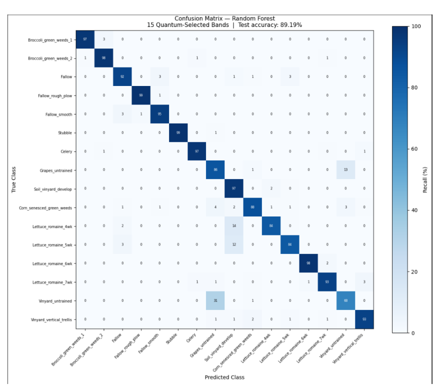
(b) IBM 15b + Random Forest
OA = 91.46%



(c) IBM 20b+sp + XGBoost
OA = 94.85%

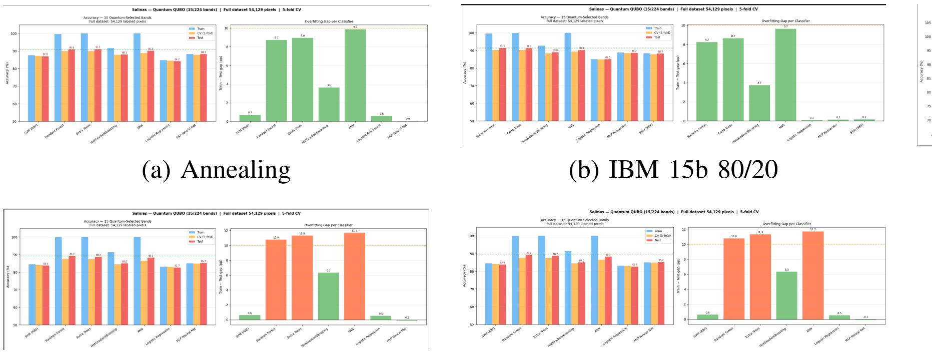


(d) SV 16q + Random Forest
OA = 89.19%



(e) SV 20q + Random Forest
OA = 89.19%

Fig. 3. Normalized confusion matrices (recall %) for the best classifier of each configuration. Config 5 (IBM, 20 bands, spatial) shows the strongest diagonal dominance. Statevector configurations (d) and (e) are identical. The primary confusion across all methods is Vinyard untrained → Grapes untrained due to spectral overlap.



(a) Annealing (b) IBM 15b 80/20 (c) IBM 20b+sp 90/10

(d) Statevector 16q (e) Statevector 20q

Fig. 4. Classifier accuracy comparison charts for all five configurations. Each chart shows Train (blue), CV 5-fold (orange), and Test (red) accuracy per classifier. The right panel in each chart shows the overfitting gap (Train − Test). Config 5 shows the highest test accuracy bars with reasonable overfitting gaps for SVM and MLP.

outperformed statevector simulation in speed (0.3 min vs. 47 min for 20q).

Band diversity, not qubit count, is the primary determinant of accuracy. The statevector 16q and 20q experiments provide a clean controlled comparison: identical QUBO formulation, identical algorithm, identical post-processing — but different candidate pool sizes (16 vs. 20 bands). The result: identical band selection (B41–55), identical QUBO cost, and identical 89.19% accuracy across all classifiers. This demonstrates conclusively that adding more qubits without expanding the candidate pool to span diverse spectral regions provides no benefit. Future work should ensure that qubit count increases are accompanied by proportionally broader candidate pools.

Simulated Annealing is a competitive classical baseline. Annealing achieves 91.06% OA (ET) — only 0.40 pp below IBM hardware (91.46%) — while being entirely classical and completing in 10.2 seconds. Its 2-cluster band selection (bands 13–16, 22–23, 41–49) is close to IBM's 3-cluster selection (bands 13–16, 22–24, 41–52 with gaps) and explains the near-competitive result. For practical applications where quantum hardware access is limited, simulated annealing with a 30-band candidate pool provides a strong alternative.

Shallow QAOA limitations and future directions. $p = 1$ QAOA is known to be suboptimal for large combinatorial problems. Despite this, IBM hardware achieved a QUBO cost within 0.12% of the classical annealing optimum, likely benefiting from the inherent exploration induced by quantum noise (effectively acting as stochastic perturbations). Deeper QAOA ($p \geq 3$) would require substantially more QPU time than available under the free-tier allocation (10-minute circuit execution limit). Future work should explore $p \geq 3$ QAOA, variational quantum eigensolvers (VQE), and quantum annealing on D-Wave hardware.

VI. Conclusion

This paper presented a comprehensive hybrid quantum-classical pipeline for hyperspectral band selection and classification on the Salinas dataset, comparing five experimental configurations spanning classical, simulated quantum, and real quantum hardware. Key conclusions are:

1) **Best overall result:** IBM `ibm_fez` with 20 quantum-selected bands, 3×3 spatial-spectral features, and XGBoost classifier achieves OA = **94.85%**, AA = **97.60%**, Kappa = **0.9426** — the highest results in this study.
2) **Spatial features provide the largest gain:** Adding 3×3 neighborhood features to 20 quantum-selected bands improved OA by +3.39 pp over the same hardware with 15 spectral-only bands.
3) **93% dimensionality reduction** (224 → 15 or 20 bands) is achieved while maintaining competitive accuracy across all seven classifiers, validating QUBO-guided selection.
4) **Band diversity is the primary driver of accuracy**, not qubit count. Statevector 16q and 20q produced identical results (89.19%) despite different qubit counts, as both candidate pools lacked cross-region diversity.
5) **IBM real hardware outperforms all simulation approaches** by 2.27 pp OA (91.46% vs. 89.19%) for spectral-only experiments, confirming the viability of NISQ-era devices for remote sensing applications.
6) **Verified IBM Quantum jobs** (IDs: `d6v7q7itnsts73estvu0` and `d70v4l0v5rlc73f6hlb0`) on `ibm_fez` confirm reproducible quantum hardware results.

Future work will explore deeper QAOA circuits ($p \geq 3$), extension to PaviaU and Indian Pines benchmarks, larger spatial windows (5×5, 7×7), and quantum-native classifiers such as quantum SVMs and quantum neural networks.

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